

Comparative genomics

Signal of selection

Medoza,
Argentina
2025



Comparative genomics using genomes

- Genome assembly
 - Reconstruction of the genome of one or more individuals
- Reference genome
 - Genome assembly that is the point representation of the structure and organisation of the genome of a species

Genome

Download a genome data package including genome, transcript and protein sequence, annotation and a data report

Selected taxa						
Homo sapiens  Enter one or more taxonomic names 						
Filters 						
Download 	Select columns	1,896 Genomes		Rows per page	20 	1-20 of 1,896  
☐ Assembly	GenBank	RefSeq	Scientific name	Modifier	Annotation	Action
☐ GRCh38.p14 	GCA_000001405.29	GCF_000001405.40	Homo sapiens (human)		NCBI RefSeq	
☐ T2T-CHM13v2.0	GCA_009914755.4	GCF_009914755.1	Homo sapiens (human)		NCBI RefSeq	
☐ hg002v1.1.mat	GCA_018852615.3		Homo sapiens (human)	NA24385 (isolate)		
☐ hg002v1.1.pat	GCA_018852605.3		Homo sapiens (human)	NA24385 (isolate)		
☐ PAN027_pat_v1.0	GCA_046332035.1		Homo sapiens (human)	HG06807 (isolate)		
☐ RPE1V1.1 Haplotype 1	GCA_050656345.1		Homo sapiens (human)	RPE-1 (isolate)		

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National Library of Medicine
National Center for Biotechnology Information

Genome assembly GRCh38.p14

reference

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Genes under selection

- Most mutations are neutral or slightly deleterious
- Most genes are under purifying selection (old genes)
- Beneficial mutations are rare
 - change in environment - local adaption
 - novel pathogens or predators
 - gene duplication events – release from purifying selection

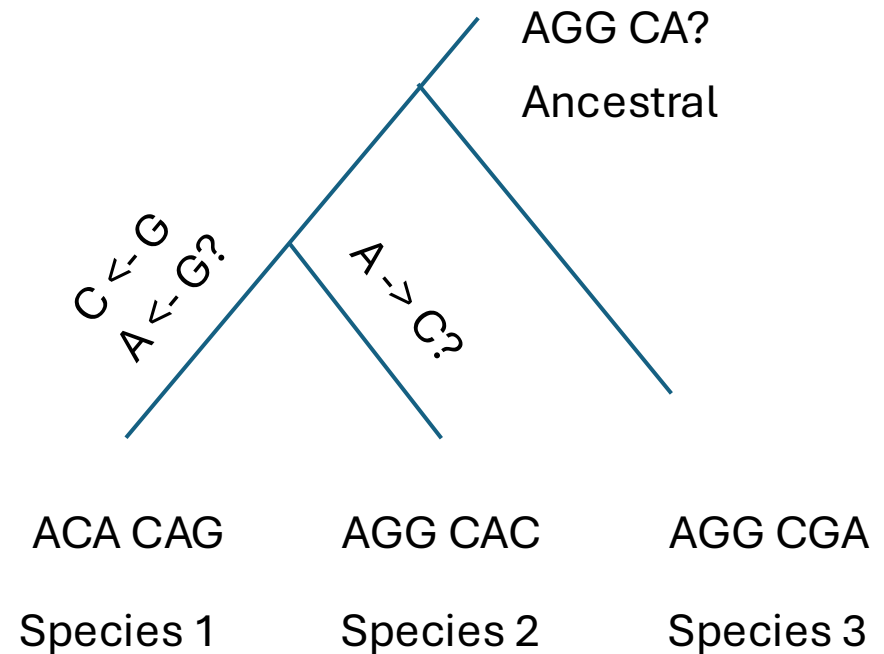
Genes under selection

- Voltage-sensitive sodium channel (para)
- Confers nerve insensitivity to DDT and pyrethroid insecticides
- Leucine to phenylalanine replacement and/or
- Methionine to threonine replacement



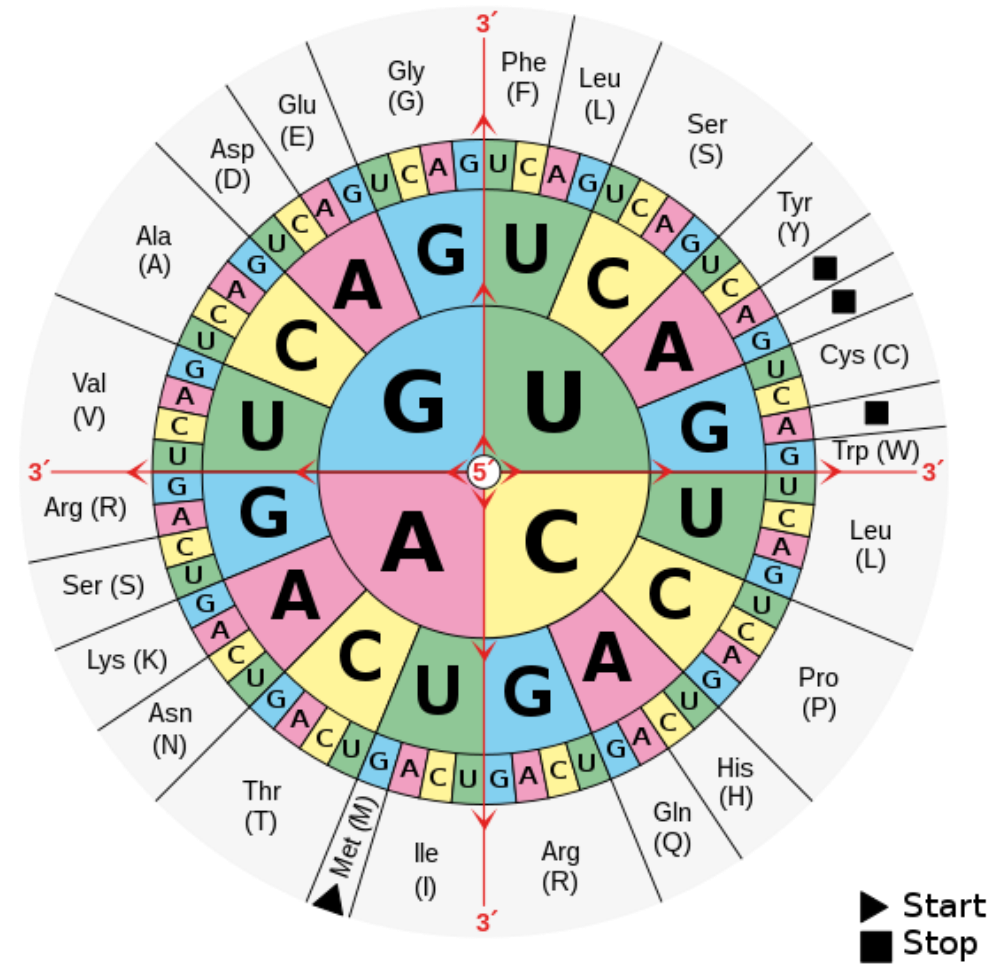
How to detect genes under selection using genome data

- Substitution-based detection in a phylogenetic framework
 - Substitutions are single nucleotide mutations that have been fixed in a species
- Ancestral state reconstruction
- Substitution rates along branches



Substitution types

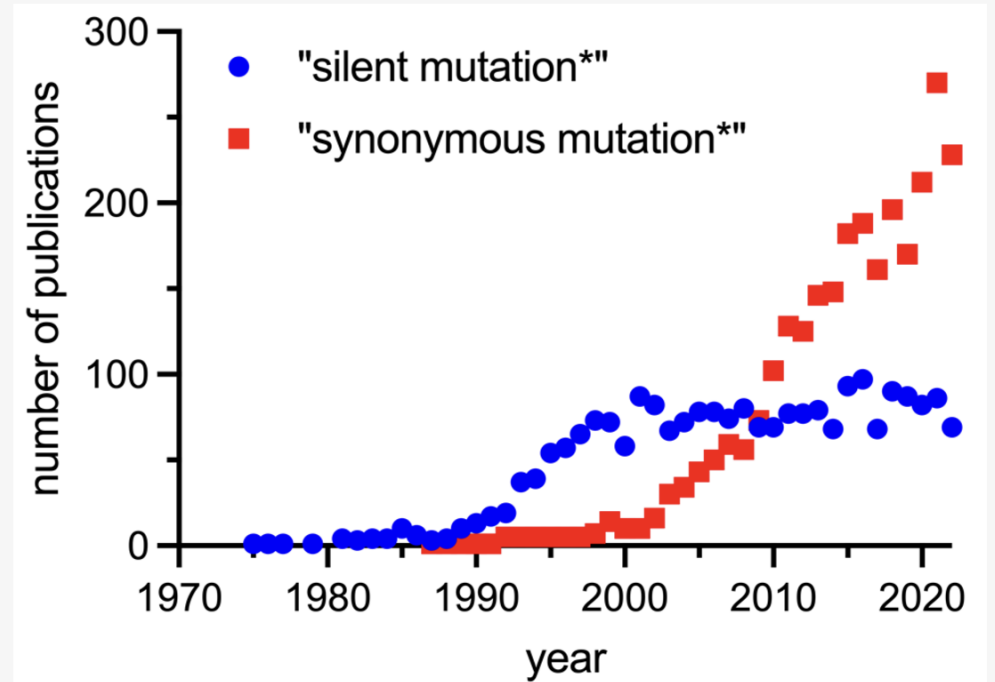
- Synonymous substitutions – no change in amino acid
- Non-synonymous substitutions
 - Change in amino acid - missense
 - From amino acid to stop codon - nonsense
 - From stop codon to amino acid - nonstop



Substitution types

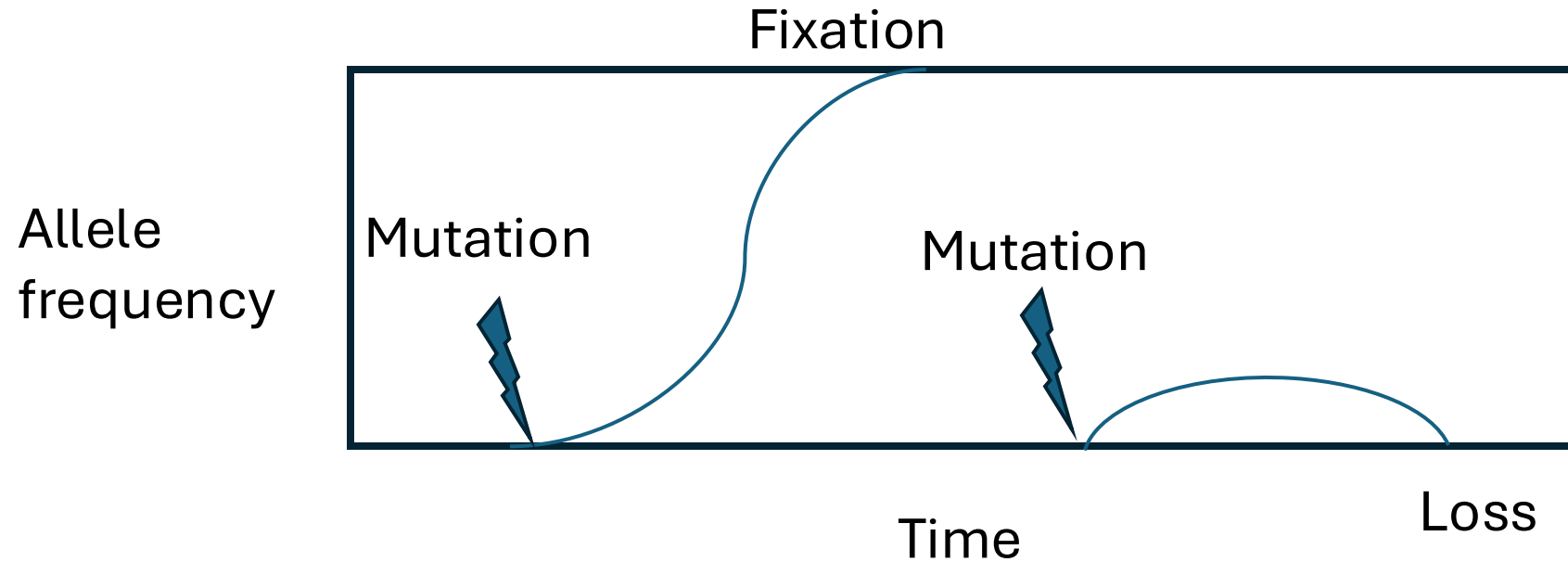
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Figure 3. Number of publications found in PubMed using the search terms "silent mutation*" and "synonymous mutation*".



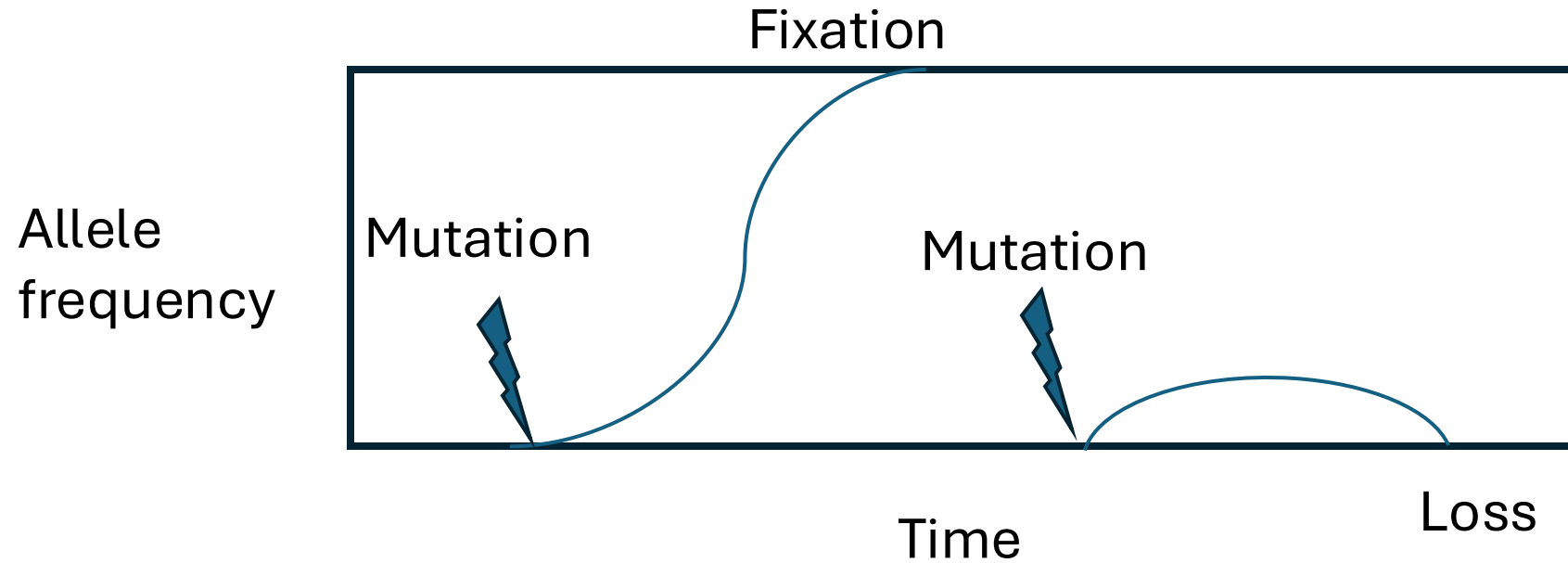
Substitutions

In neutral positions the loss or fixation of a mutations is due to drift



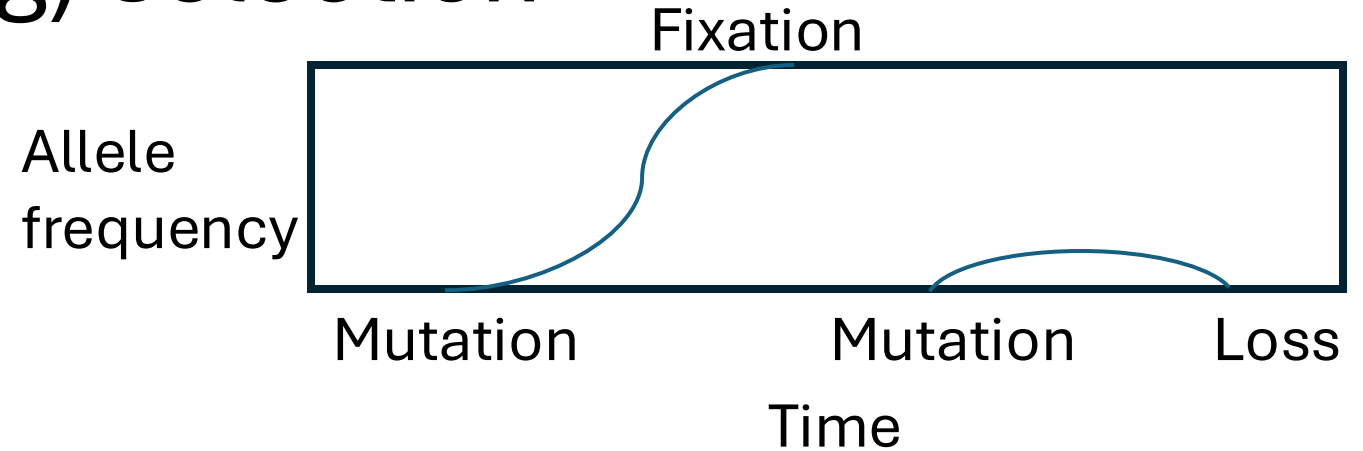
Substitutions

Synonymous substitutions

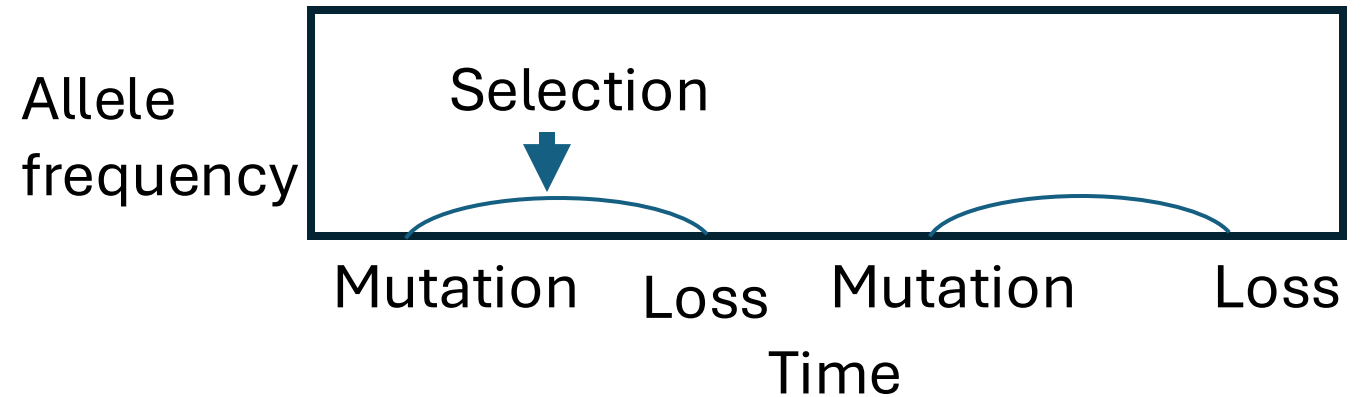


Negative (purifying) selection

Synonymous substitutions

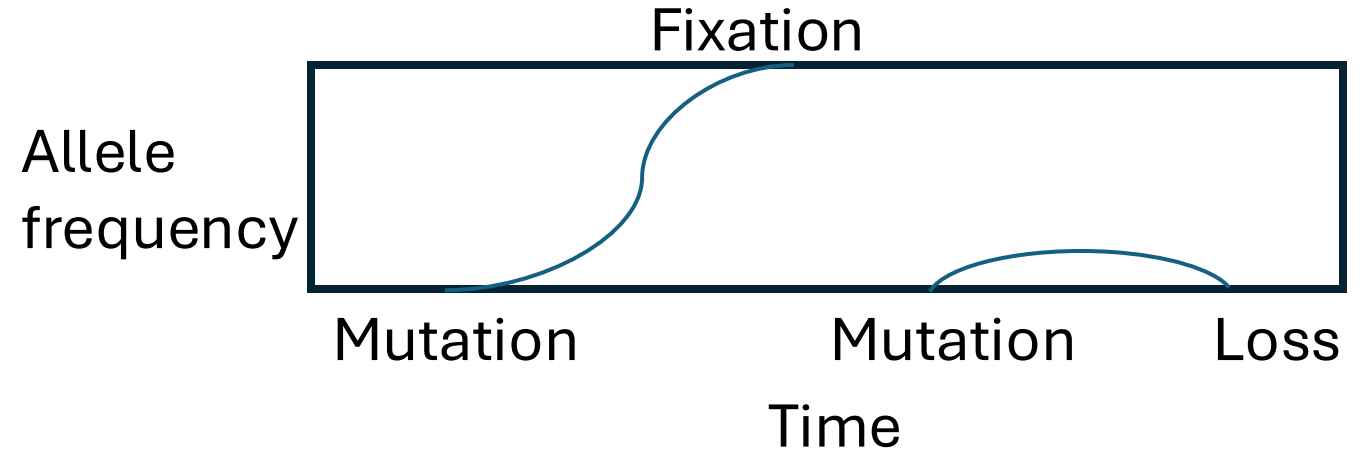


Non-synonymous substitutions

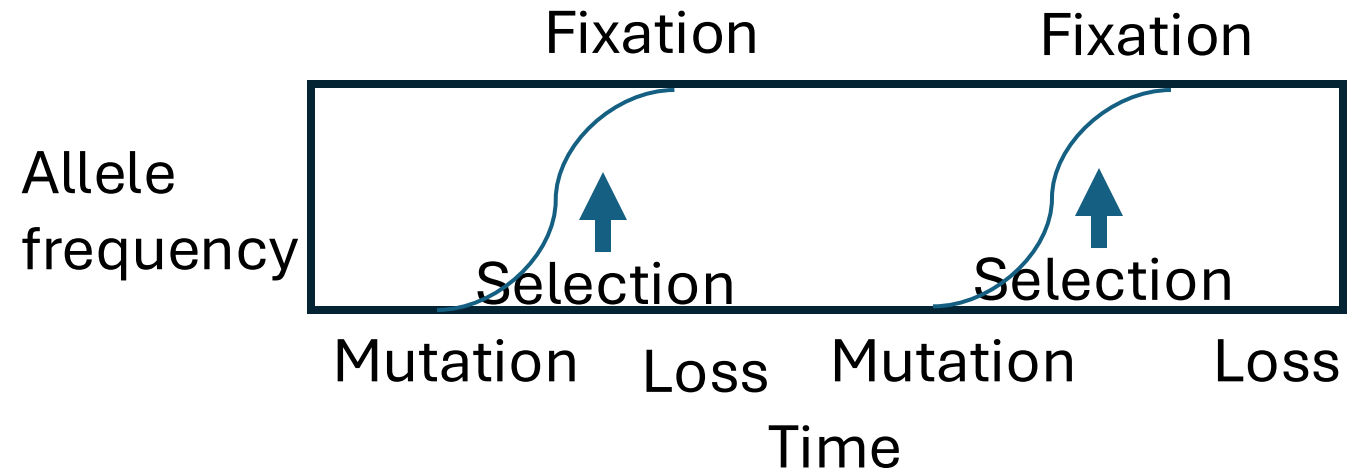


Positive selection

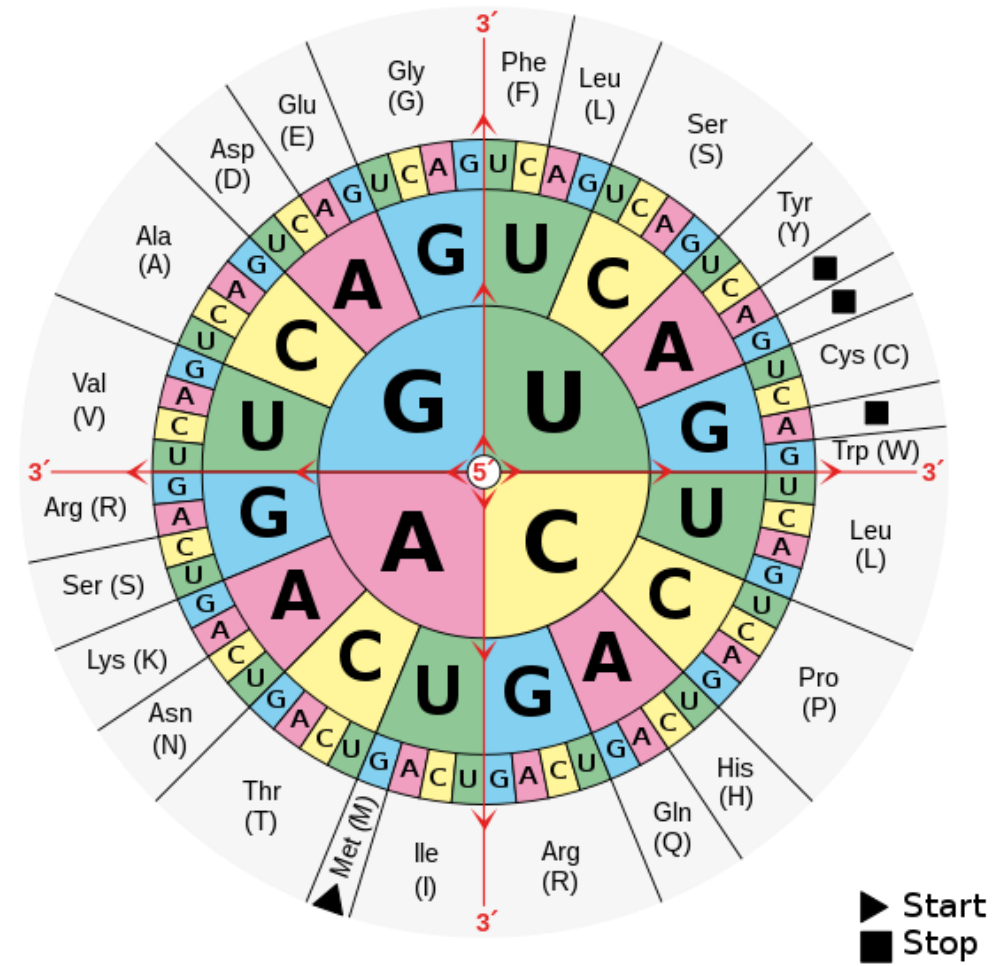
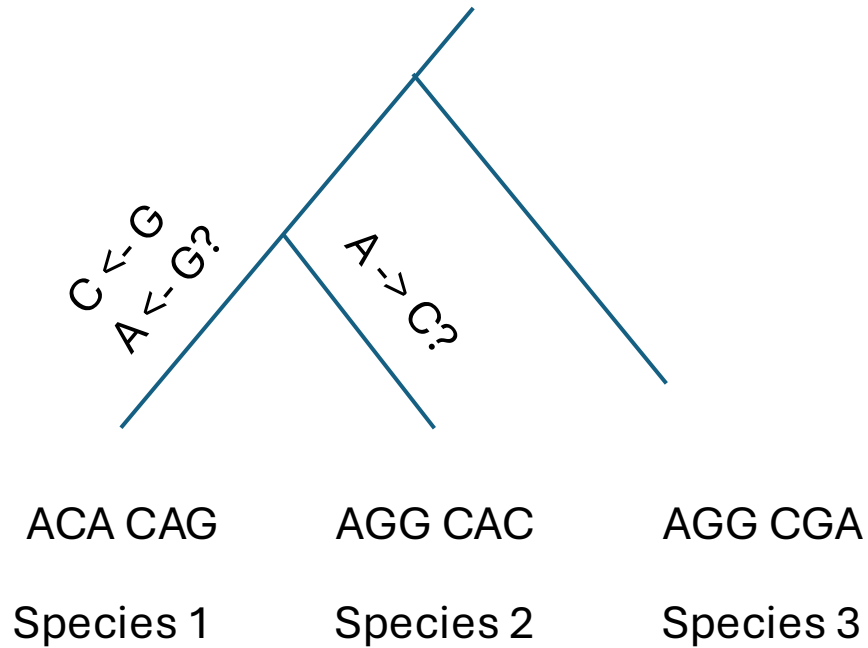
Synonymous substitutions



Non-synonymous substitutions

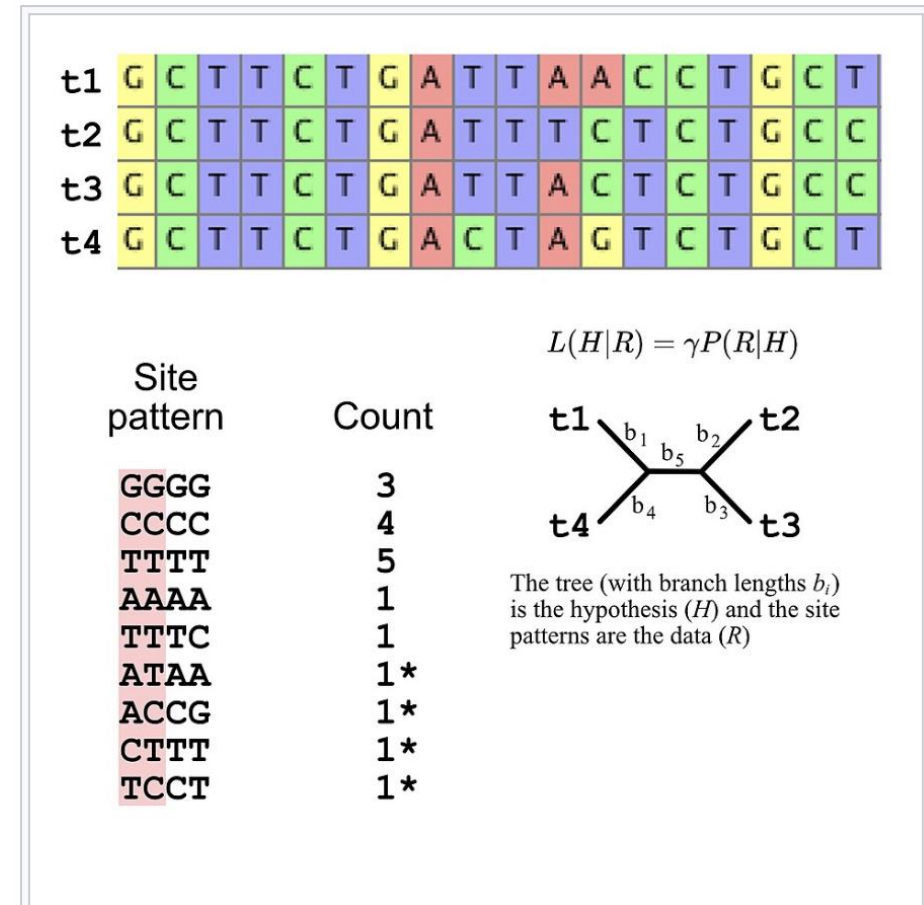


How do we count the number of substitutions?



Infer evolutionary rate with substitution models

- Multiple sequence alignment (inferred using substitutions models)
- The data converted to a set of site patterns
- The site patterns are used to calculate the likelihood with a
 - substitution model
 - phylogenetic tree
- The substitution model is used to estimate evolutionary distances for pairs of sequence
 - Distances are the number of substitutions that have occurred since the sequences had a common ancestor
- The evolutionary distance equation (Jukes and Cantor, 1969).
 - Transforms the proportion of nucleotide differences into evolutionary distance



Substitution models

- Substitution models
 - observed differences -> actual number of substitutions during the evolutionary history

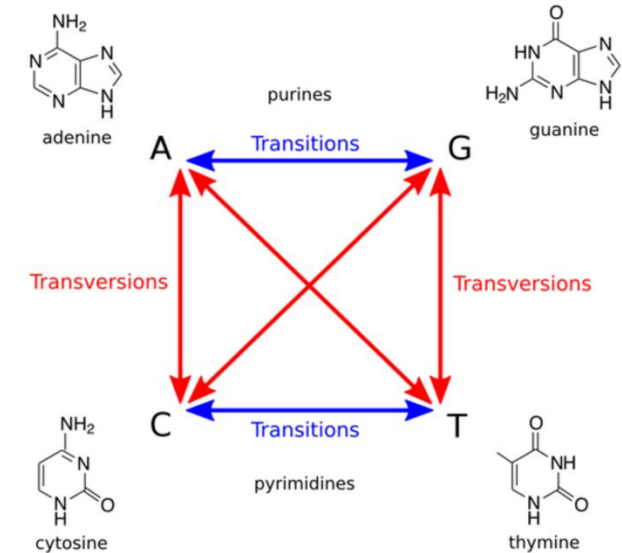
Substitution matrix

$$M_{ij} = \begin{matrix} & \begin{matrix} A & C & G & T \end{matrix} \\ \begin{pmatrix} 1-3\epsilon & \epsilon & \epsilon & \epsilon \\ \epsilon & 1-3\epsilon & \epsilon & \epsilon \\ \epsilon & \epsilon & 1-3\epsilon & \epsilon \\ \epsilon & \epsilon & \epsilon & 1-3\epsilon \end{pmatrix} & \begin{matrix} A \\ C \\ G \\ T \end{matrix} \end{matrix}$$

A-C, A-G, A-T, C-G, C-T and G-T

Substitution models, examples

JC or JC69	0	Equal substitution rates and equal base frequencies (Jukes and Cantor, 1969).
F81	3	Equal rates but unequal base freq. (Felsenstein, 1981).
K80 or K2P	1	Unequal transition/transversion rates and equal base freq. (Kimura, 1980).
HKY or HKY85	4	Unequal transition/transversion rates and unequal base freq. (Hasegawa, Kishino and Yano, 1985).
TN or TN93	5	Like HKY but unequal purine/pyrimidine rates (Tamura and Nei, 1993).
TNe	2	Like TN but equal base freq.
K81 or K3P	2	Three substitution types model and equal base freq. (Kimura, 1981).
K81u	5	Like K81 but unequal base freq.
GTR	8	General time reversal model, unequal rates and unequal base freq. (Tavare, 1986).



$$M_{ij} = \begin{pmatrix} 1 - 3\epsilon & \epsilon & \epsilon & \epsilon \\ \epsilon & 1 - 3\epsilon & \epsilon & \epsilon \\ \epsilon & \epsilon & 1 - 3\epsilon & \epsilon \\ \epsilon & \epsilon & \epsilon & 1 - 3\epsilon \end{pmatrix}$$

A-C, A-G, A-T, C-G, C-T and G-T

- +F base frequencies
- +I invariable sites
- +G gamma variation

Inference of selection - d_N/d_S

- Genome from one individual per species can be used
- Number of substitutions per synonymous site (d_S)
- Number of substitutions per non-synonymous site (d_N)
- Usually inferred per gene or window-based over multiple genes

Inference of selection - d_N/d_S

- Under neutrality: $d_N = d_S$
- Purifying selection: $d_N/d_S < 1$
- Positive selection: $d_N/d_S > 1$

Inference of selection - d_N/d_S

- Codon alignment of coding nucleotide sequences
- Medium divergence
 - total branch length > 0.5 (d_S)
 - Recently diverged –low number of fixed differences,
 - Very distant – substitution saturation (recurrent mutations, problem with alignment)
- codeml – model comparison detecting positive selection using codon substitution models
- mapNH – time heterogeneous models, account for non-stationary GC-content

codeml

Table 3
Parameters in Branch-Site Model A

Site class	Proportion of sites	Background ω	Foreground ω
0	p_0	$0 < \omega_0 < 1$	$0 < \omega_0 < 1$
1	p_1	$\omega_1 = 1$	$\omega_1 = 1$
2a	$(1 - p_0 - p_1) \cdot$ $p_0 / (p_0 + p_1)$	$0 < \omega_0 < 1$	$\omega_2 \geq 1$
2b	$(1 - p_0 - p_1) \cdot$ $p_1 / (p_0 + p_1)$	$\omega_1 = 1$	$\omega_2 \geq 1$

NOTE.—Model A is the alternative hypothesis for the branch-site test of positive selection. The null model fixes $\omega_2 = 1$ for the foreground branch.

Exercise

- Use BUSCO genes (already prepared)
- Cluster orthogroups and select single copy orthologs - Orthofinder
- Alignment - we need to align homologous sites - PRANK
- Phylogeny reconstruction, tree building - IQ-tree
- Infer evolutionary rates and signs of selection - codeml